

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 100
timesteps_orig: 60
timesteps: 1
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.1
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 20

LSTM Multi-Step Training Report

Report Generated: 2025-05-10 14:39:04

Training Start: 2025-05-10_14-14-35
Training End: 2025-05-10_14-15-41
Training Duration: 1.10 minutes

Best Hyperparameters:
Number of Layers: 2
Activation: relu
Optimizer: rmsprop
Dropout Rate: 0.10

MAE: 28133.1270
MSE: 1235010077.9159
RMSE: 35142.7102
R2 Score: 0.7290

Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0355
MSE: 0.0019
RMSE: 0.0434
R2 Score: 0.8120

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 10,265 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

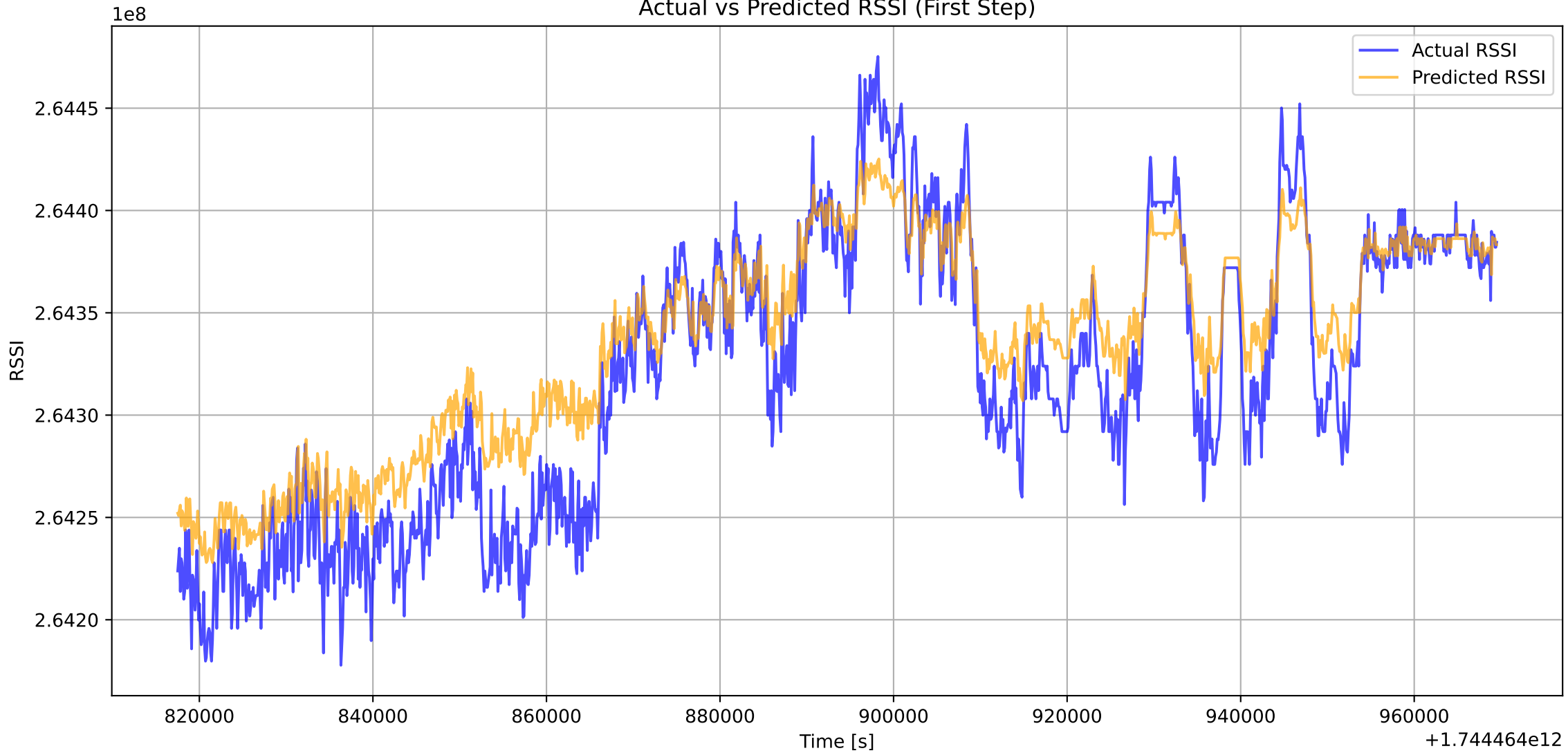
1744397085760
1744397085801
1744397086014
1744397086103
1744397086208

...

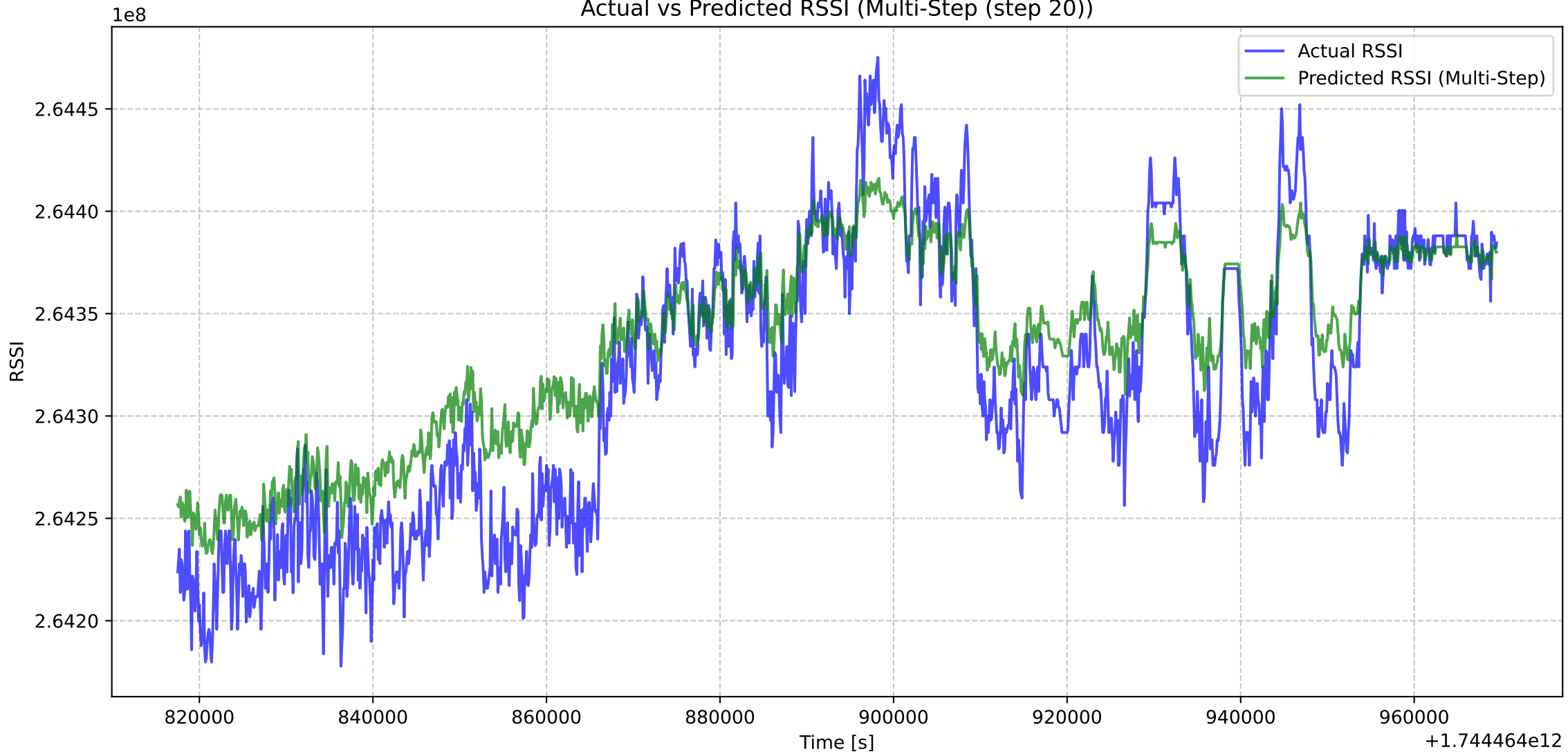
Last 5 timestamps AFTER:

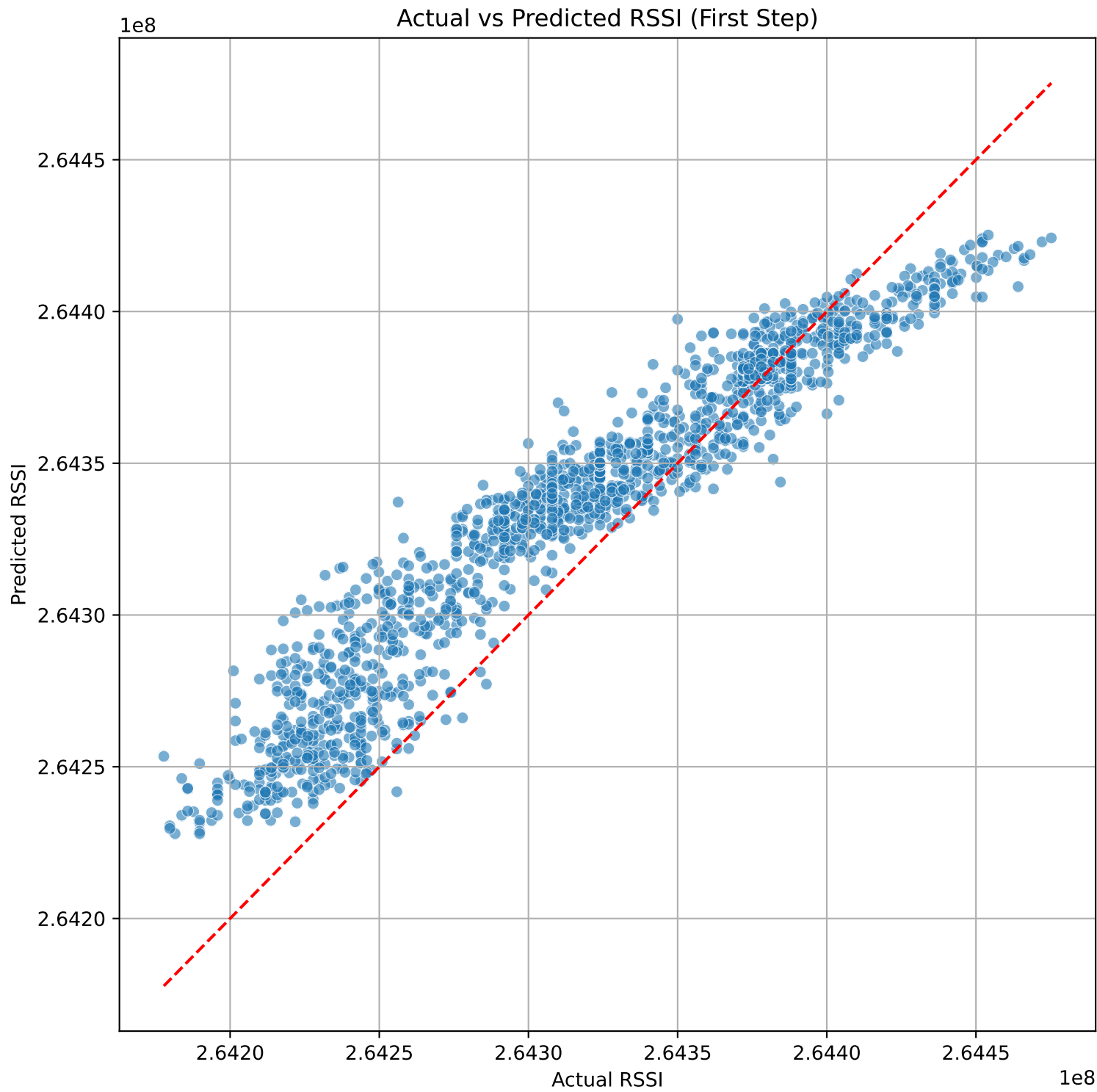
1744464971004
1744464971101
1744464971209
1744464971306
1744464971402

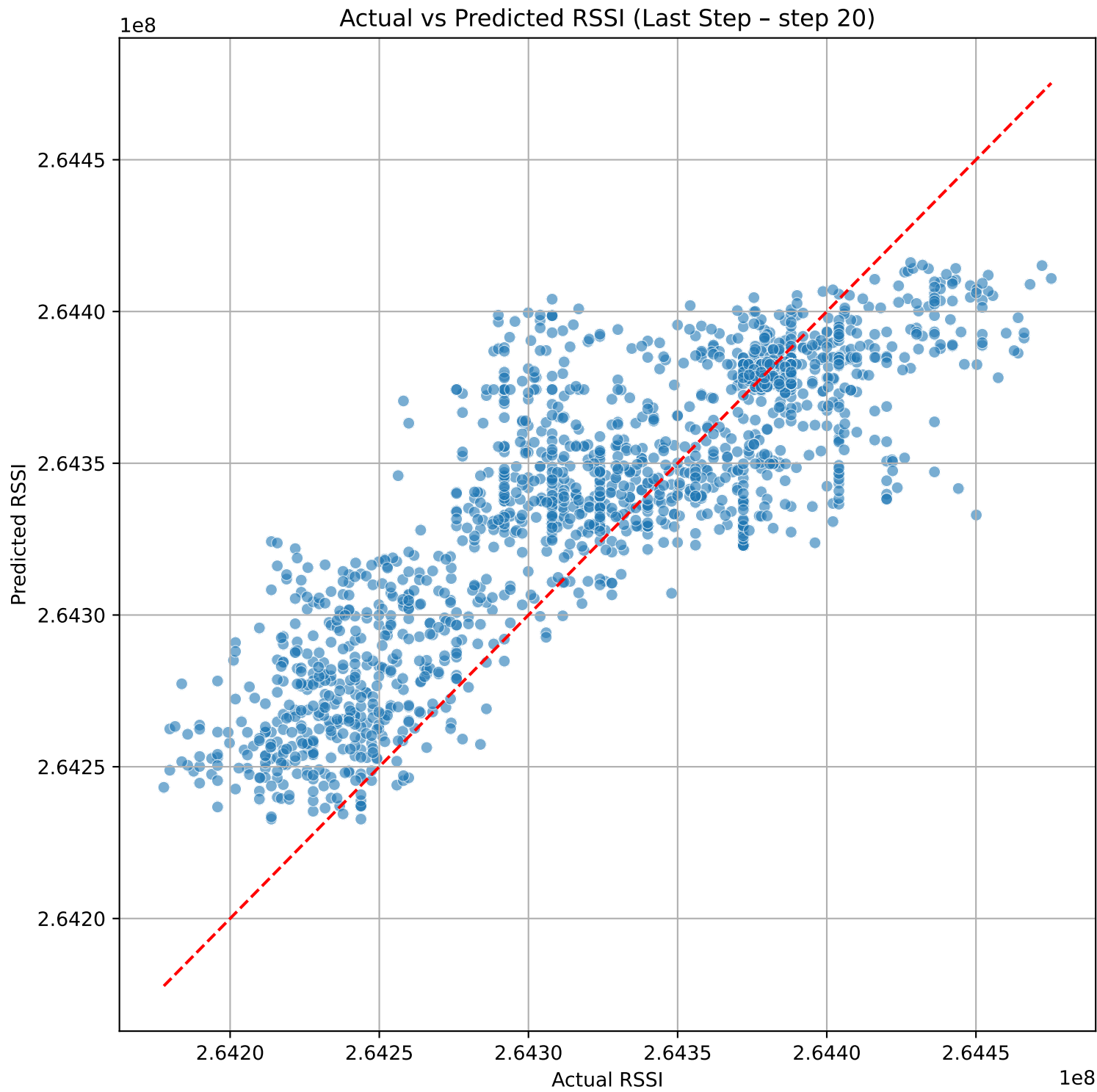
Actual vs Predicted RSSI (First Step)



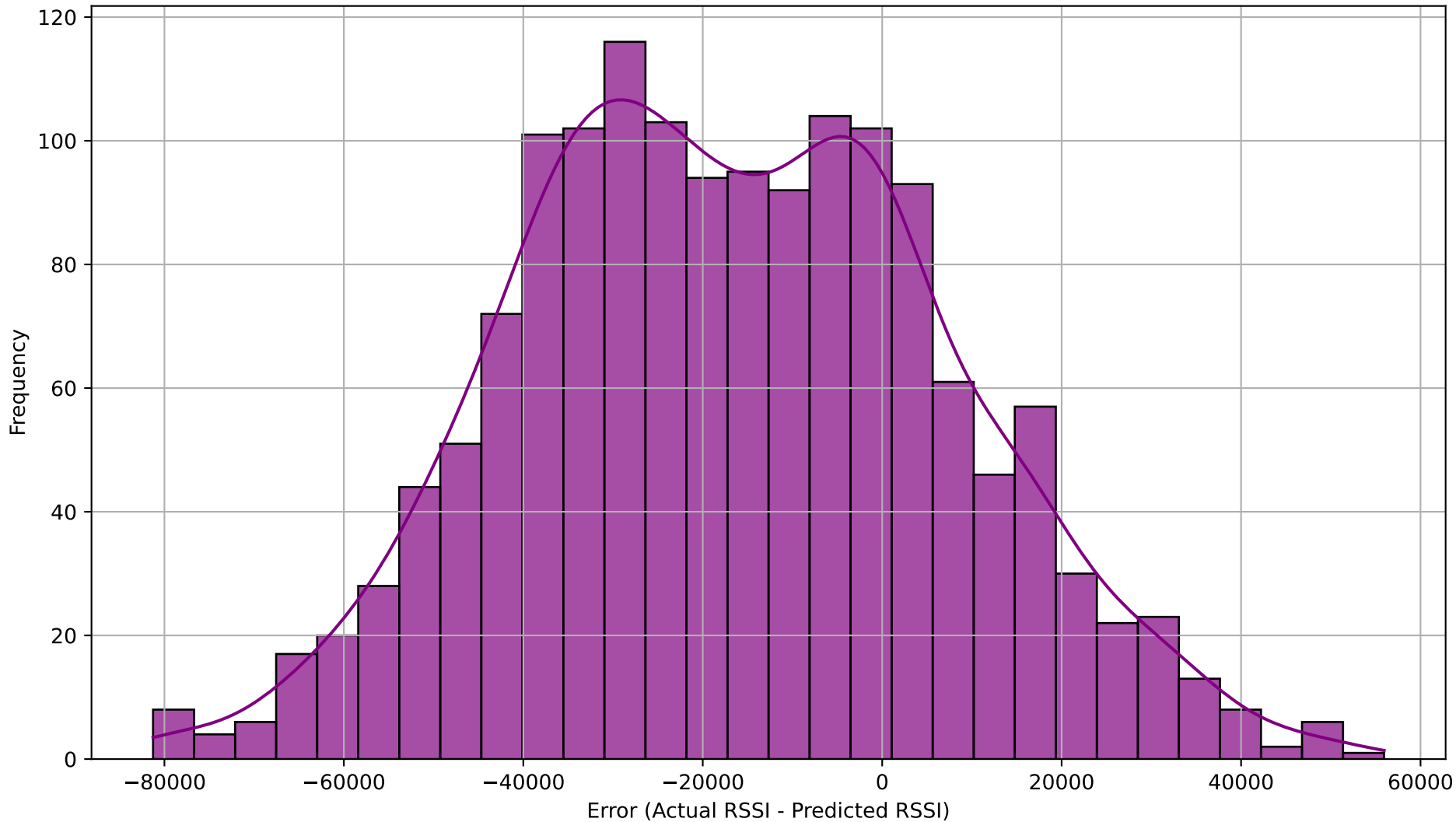
Actual vs Predicted RSSI (Multi-Step (step 20))



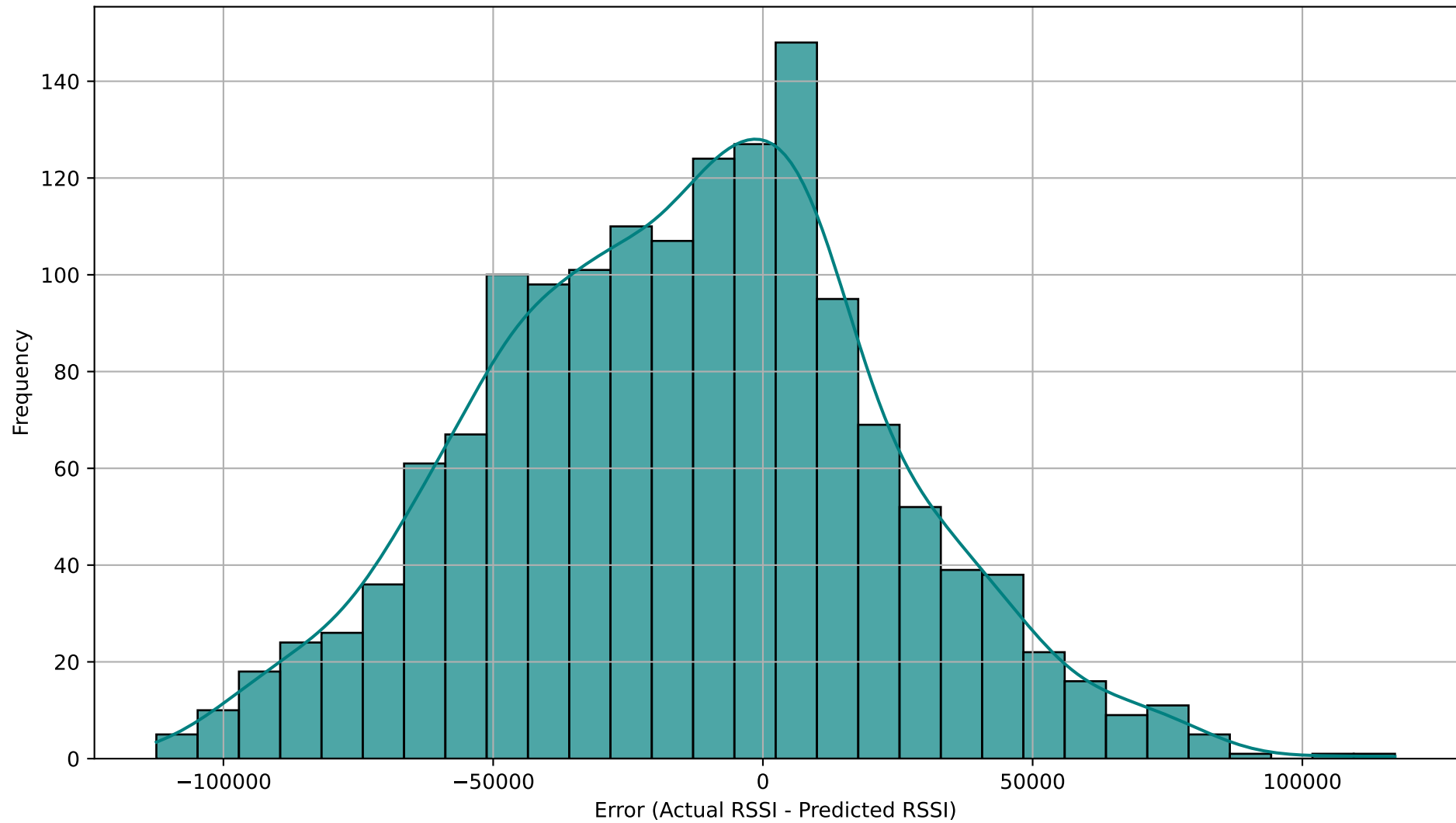




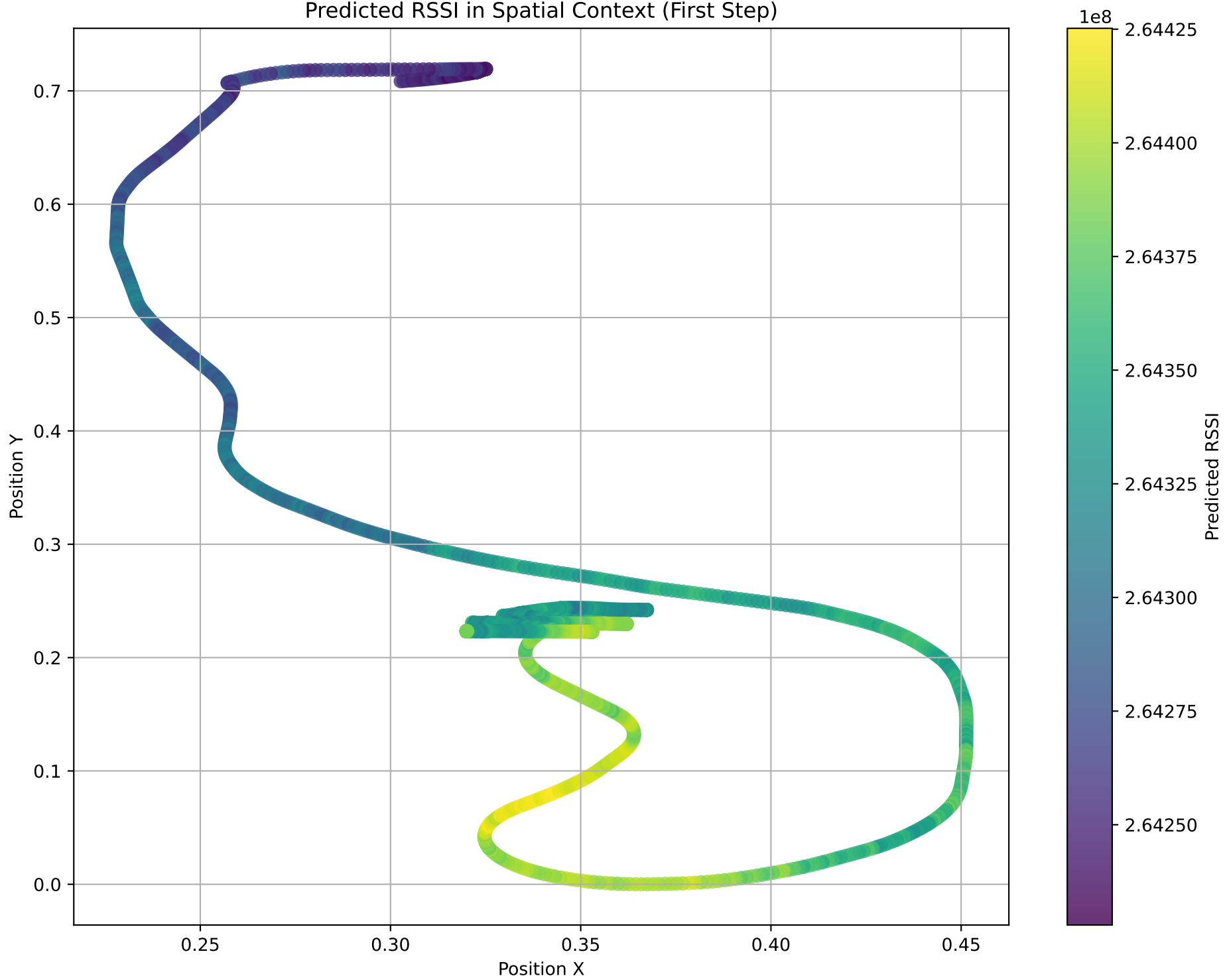
Distribution of Prediction Errors (First Step)



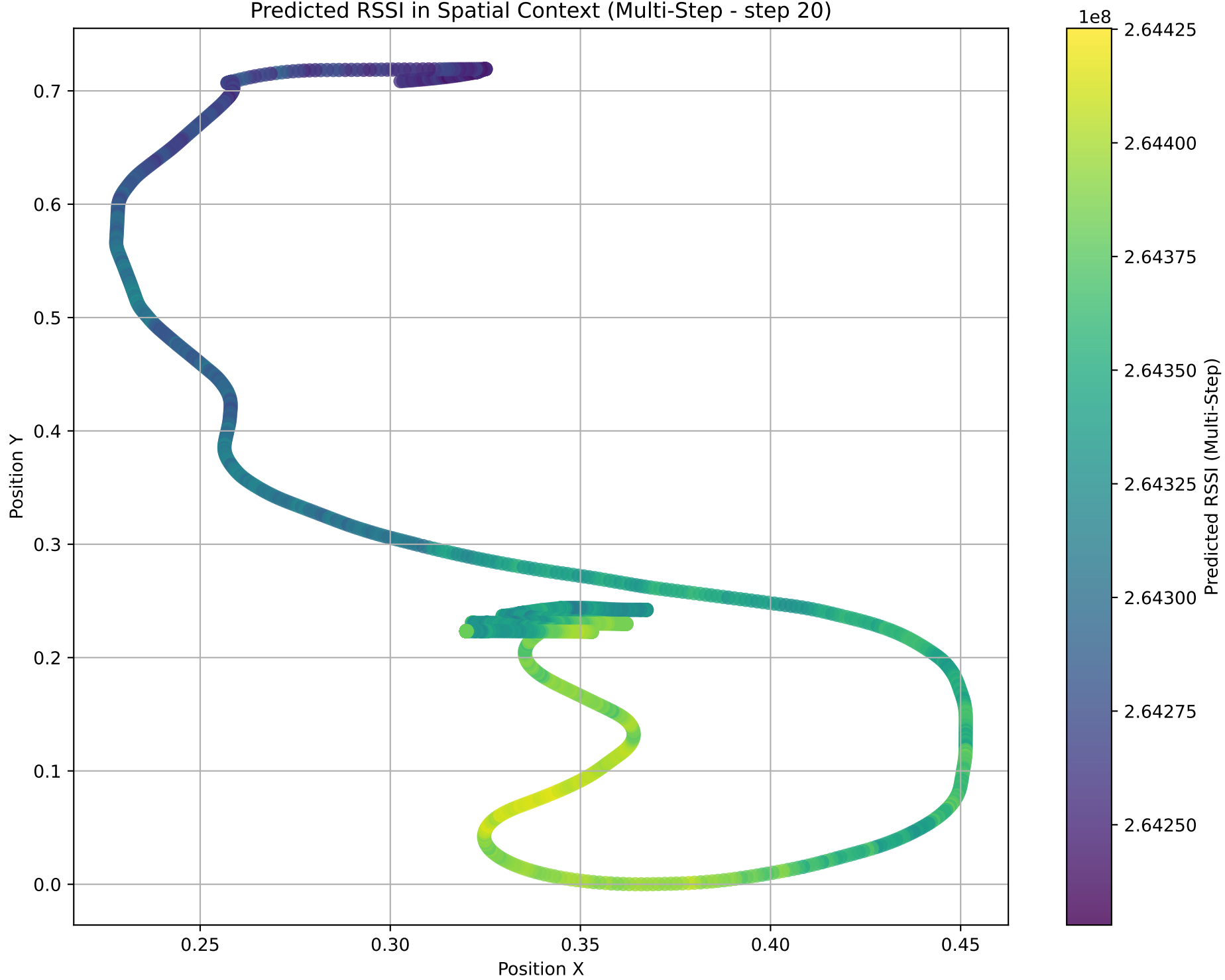
Distribution of Prediction Errors (Last Step - step 20)



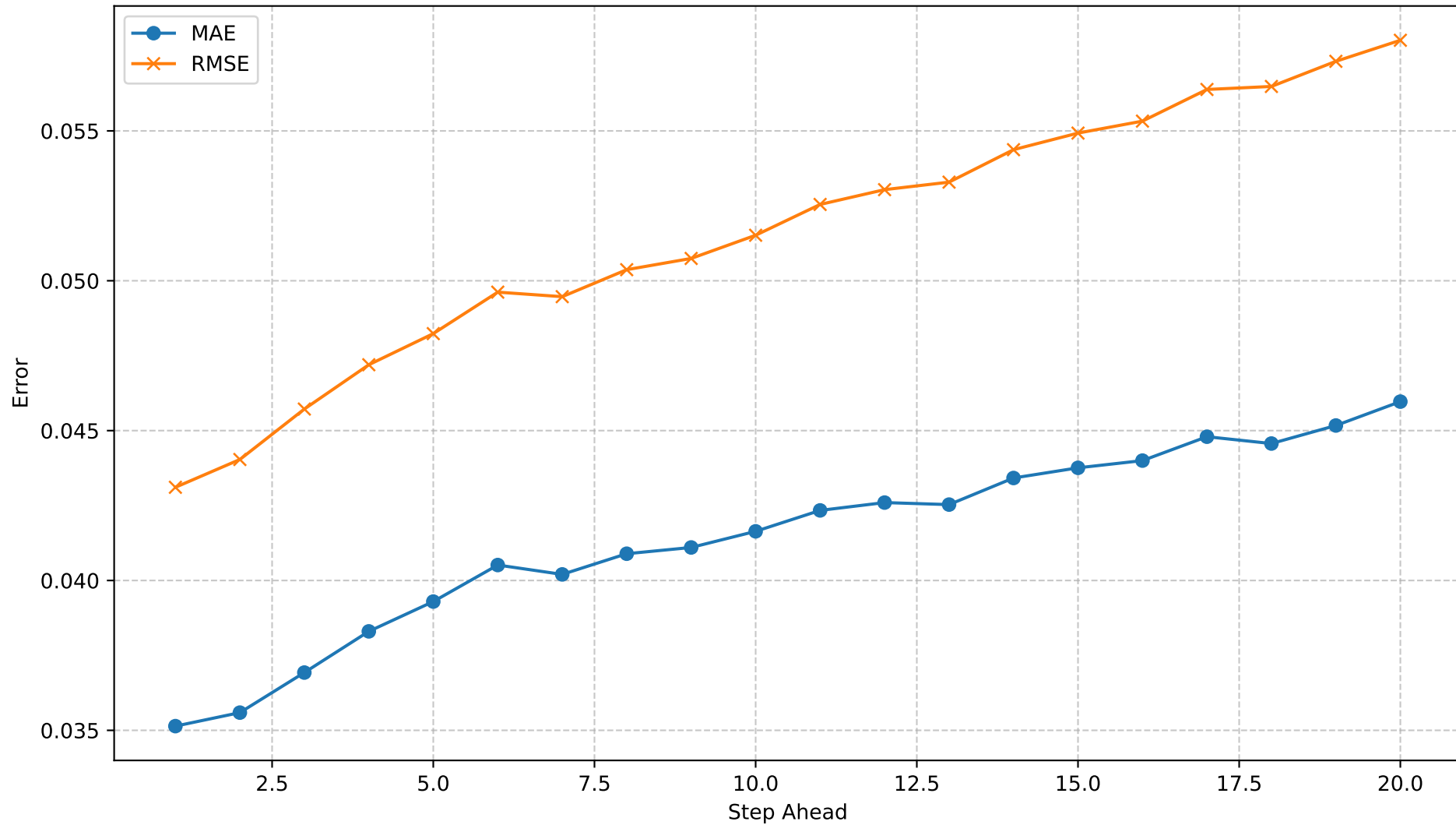
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.04 / 0.04
- Mediana : 0.04 / 0.05
- 20º passo : 0.05 / 0.06

Crescimento do erro: 30.8% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 20 (0.05), 19 (0.05), 17 (0.04)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.